Group 4

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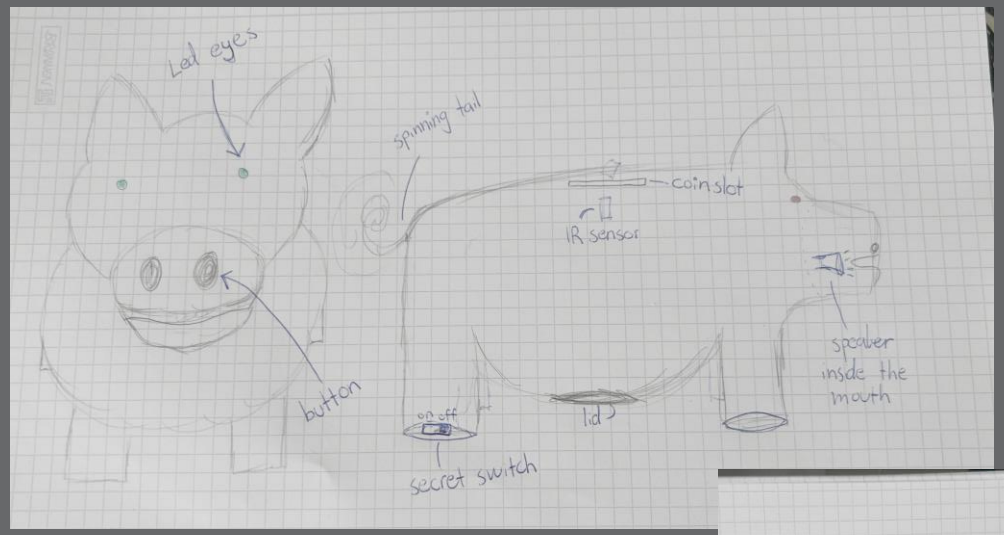
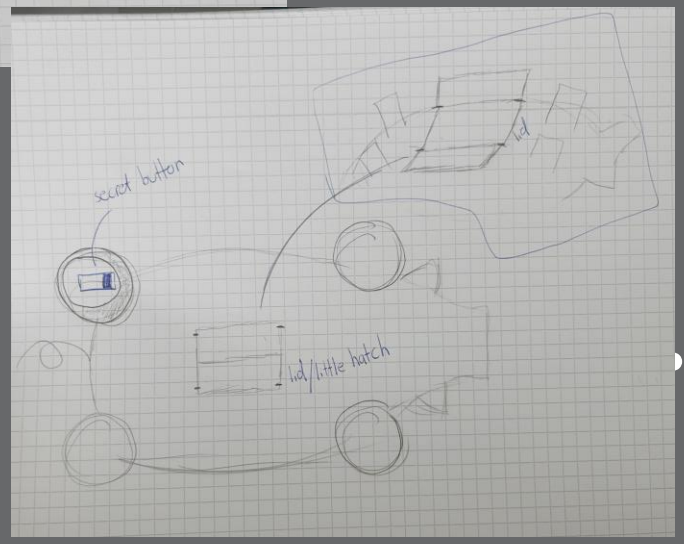
Expressive piggy bank





Description

- When you put money inside of the piggy bank it's eyes turn green, it makes a happy sound, the tail spins, and the nose twitches.
- If the secret switch is on and you press a button to open it's lid, it's eyes turn red, it makes an angry sound and the lid doesn't open.
- If the secret switch is of and you press a button to open it's lid, you can open the lid safely.

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Expressive piggy bank

Parts

Electronics: Raspberry Pi Pico, breadboard, jumper wires, resistors, IR sensor, push button, switch, LEDS, buzzer, (2-3) servo motors, USB cable for microcontroller, speaker

Non-electronics: PLA filament for 3D printer, hinges for lid, rubber bands, screws (possibly), glue.

Sustainability

- *We'll design it in a way that makes it easily repairable.*
- *We'll 3D print the piggy bank in atleast 4 parts (modular design).*

Challenges

- *Making the IR-sensor work properly.*
- *Some of us have to do tasks that we're not so experienced with.*
- *Unusual shape might make fitting the components difficult. (3D-design critical)*